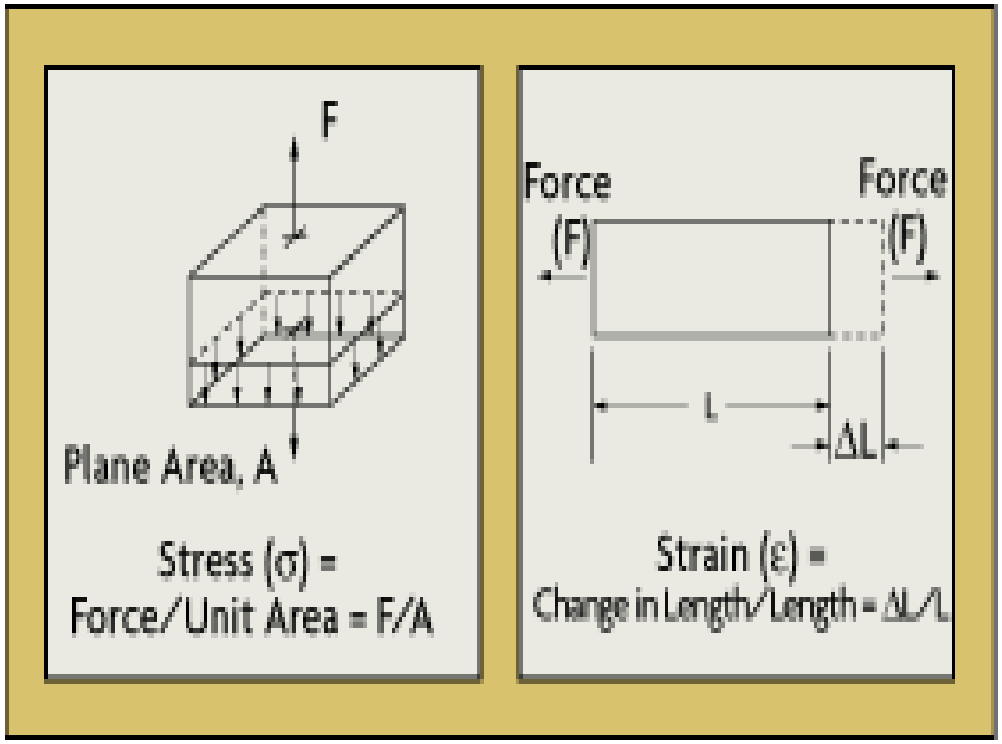
STRAIN GAUGES

What is Mechanical Strain?

- If a material with an area **A**, receives a tensile force **P**, a stress σ will be produced relative to that force
- Stress is defined as the object's internal resisting forces $$\sigma = \frac{F}{A}$$
- The materials cross section contracts and its length elongates by ΔL from the original length L



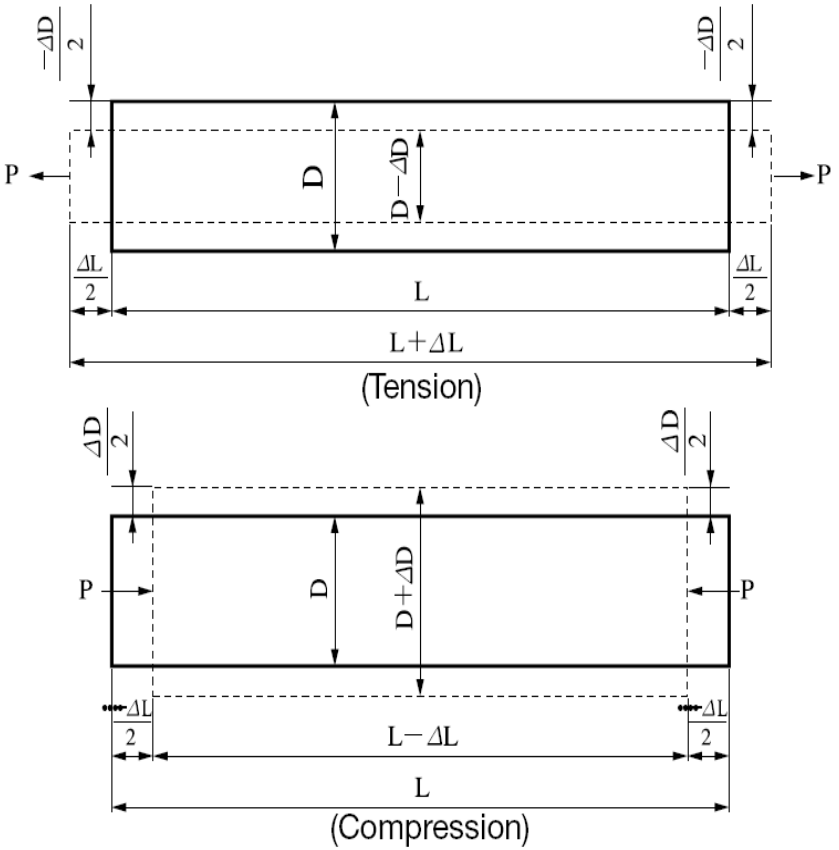
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Mechanical Strain (cont)

- This ratio is known as the tensile strain and is expressed as

$$\varepsilon = \frac{\Delta L}{L} \quad [\mu\varepsilon]$$

- Strain is defined as the displacement and deformation that occur.



What is Relationship Between Stress and Strain?

- Stress is related to strain by Hooke's Law

$$\sigma = E\varepsilon$$

- Where **E** is the Young's Modulus of the material

- The ratio of longitudinal strain (elongation in the axial direction) and transverse strain (contraction in the transverse direction) is known as Poisson's Ratio

- Where ε_1 is the longitudinal strain and ε_2 is the transverse strain

$$\nu = \left| \frac{\varepsilon_2}{\varepsilon_1} \right|$$

Strain Gauges

- Many physical variables can be determined from the amount of strain on a system

- A strain gauge, in its simplest form is a piece of wire which is attached to the object under strain. If the object changes its dimensions so will the piece of wire. This will change the resistance of the wire

- Strain gauges are made in three main types:
  1) foil strain gauges,
  2) wire strain gauges,
  3) semiconductor strain gauges

Strain Gauges

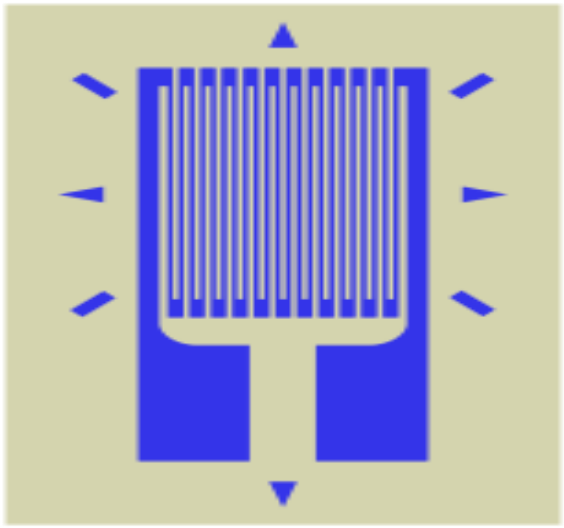
- Many physical variables can be translated into strain
- Therefore if we measure the strain in a system, we can measure those variables
- For example we can measure the pressure in a pressure vessel by attaching strain gauges to the vessel walls

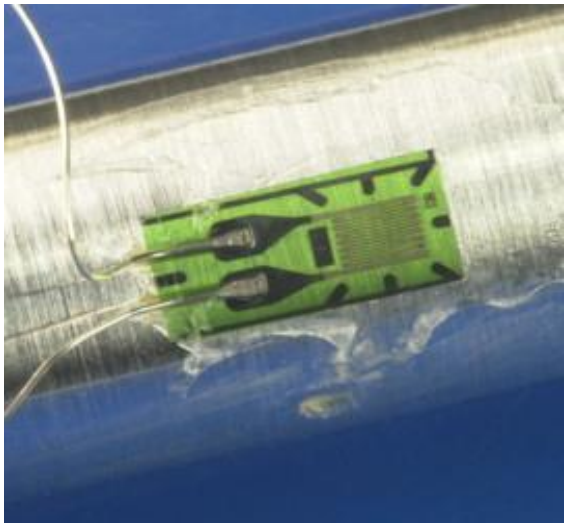


Kvaerner - Ammada Hess - Triton
Hydrocyclone & Downstream Enhancement Vessels

KGD Industrial Services

What is a Strain Gauge?





- It is a form of resistive transducer
- The primary measurand in a resistive transducer either directly or indirectly causes a change in ohmic resistance of an electrical element
- In its simplest form it is a wire attached to the object under test
- If the object changes its dimensions, so with the strain gauge

The Resistivity Equation

- The resistance of a substance is defined by its length (***l***), cross sectional area (***A***) and resistivity (ρ)

$$R = \frac{\rho l}{A}$$

- Resistivity is a property of a material

- A change in any one of these factors will result in a change the resistance of the material

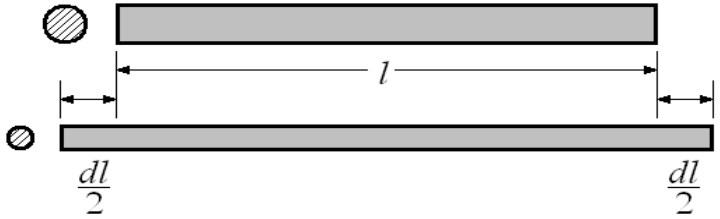
The Resistivity Equation

- Resistivity is also temperature dependent
- The resistance variations with temperature is given by

$$R_{T2} = R_{T1}(1+\alpha T)$$

- Where R_{T2} is the resistance at temperature T2, R_{T1} is the resistance at temperature T1, T is difference between T1 ands T2, α is the temperature coefficient.
- Some materials have wide linear temperature range (perfect to be temperature sensors)

The Sensitivity Factor



- If we stretch a wire, its length will increase, the resistivity will alter slightly depending on the material and the cross sectional area will be reduced
- The sensitivity of the strain gauge is sometimes expressed as Gauge factor

$$G = \frac{\Delta R / R}{\Delta L / L}$$

- G=relative change in resistance due to the change in relative length
- The change in resistance is $$dR = \frac{\rho dl}{A} - \frac{\rho l dA}{A^2} + \frac{l d\rho}{A}$$
- So that $$\frac{dR}{R} = \frac{dl}{l} - \frac{dA}{A} + \frac{d\rho}{\rho}$$

The Sensitivity Factor(*Cont.*)

- How much does the area change by?
- The change in the length and area are related by Poisson's Ratio. If we consider a wire with a radius of ***r*** then

$$\varepsilon_r = -\nu\varepsilon_1 = -\nu\frac{dl}{l}$$

- The rate of change of cross sectional area is related to the square of the linear change in the radius

$$dA = A - A_0 = \pi(r + \Delta r)^2 - \pi r^2$$

$$\frac{dA}{A} = (1+\varepsilon_r)^2 - 1 = 2\varepsilon_r + \varepsilon_r^2 \approx 2\varepsilon_r = -2\nu\frac{dl}{l}$$

The Sensitivity Factor (Cont.)

- We can then rewrite the change in resistance as

$$\frac{dR}{R} = \frac{dl}{l} + 2\nu\frac{dl}{l} + \frac{d\rho}{\rho} = (1+2\nu)\varepsilon_1 + \frac{d\rho}{\rho}$$

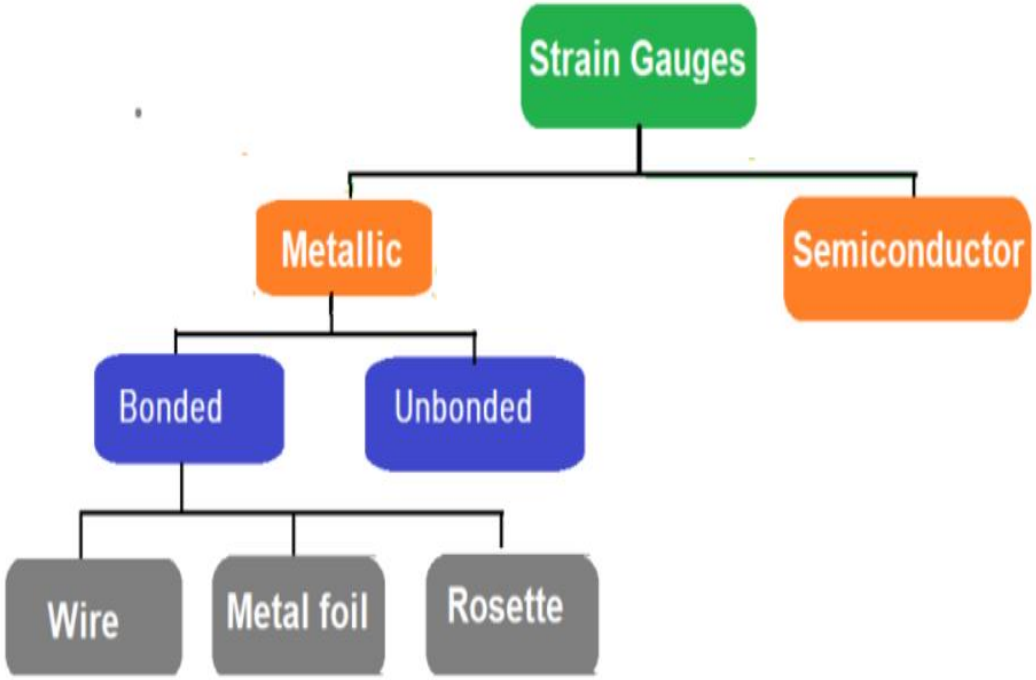
- This means that we can define a gauge factor or sensitivity factor for any type any type of strain gauge where

$$S = \frac{dR/R}{\varepsilon_1} = 1 + 2\nu + \frac{d\rho/\rho}{\varepsilon_1}$$

- We can then calculate the strain using the sensitivity factor of the strain gauge

$$\varepsilon_1 = \frac{\Delta R}{SR}$$

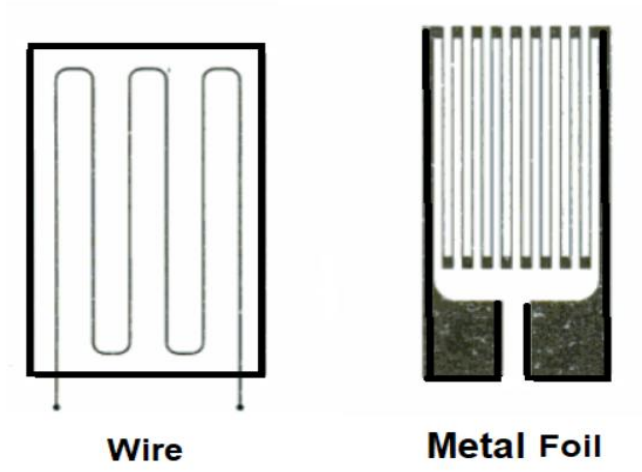
Strain Gauges Types and Classifications

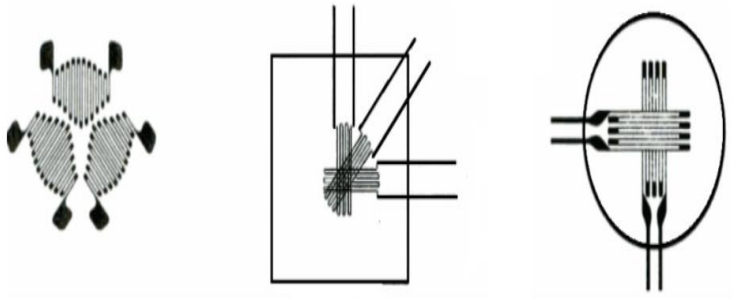


Types of Metalic Strain Gauges:

Bonded Strain Gauges

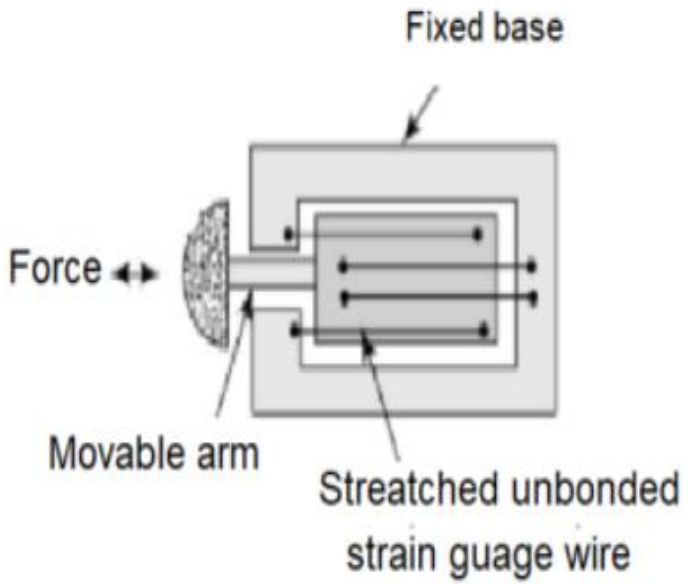
- These are the most common types of strain gauges
- They consist of either of foil or wire which has been stuck on to a bonding material
- The gauge is stuck onto a material to which stress is applied. The resulting strain is communicated to the gauge via the bonding material
- Examples of bonded strain gauges are:
  - foil gauges,
  - wire gauges,
  - rosette





Various types of Rosette strain gauges

Types of Strain Gauges: Unbonded Strain Gauges

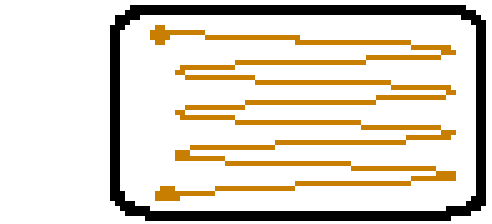


Mansfield, P.H. (1973) Electrical Transducers for Industrial Measurement

- This consists of a wire stretched between two points in an insulating medium such as air
- Measure very small motion(micrometers)
- This is not a very common arrangement since it is less robust and also there is a lot of improvement in the bonding materials
- It is usually wired in a Full Wheatstone Bridge arrangement so that two of the gauges will be lengthened while the other two will be shortened by the displacement of the movable part

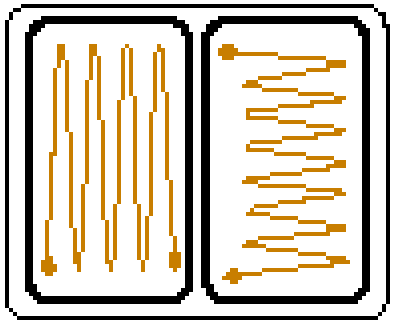
Typical Gauge Patterns: Uniaxial Strain Gauge

- The gauge pattern refers to layout and number of the grid
- Select a uniaxial strain gauge if:
  - Only one direction of strain needs to be measured
  - Cost is an issue



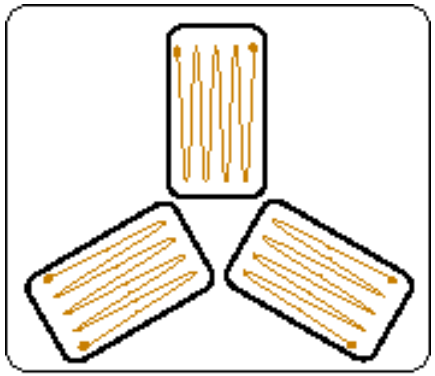
Uniaxial Strain Gage

Typical Gauge Patterns: Bi-axial and Tri-Axial Strain Rosettes



0°-90° T Rosette

- Select a bi-axial strain rosette (0º -90º Tee Rosette) if:
  - The principle stresses need to be investigated and the principle axes are known

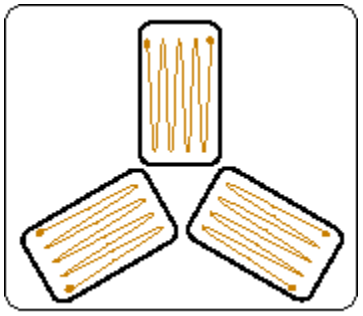


0°-60°-120° Delta Rosette

- Select a tri-axial strain rosette (0º -45º- 90º or 0º-60º-120º Delta Rosette) if:
  - The principle stresses need to be investigated and the principle axes are unknown

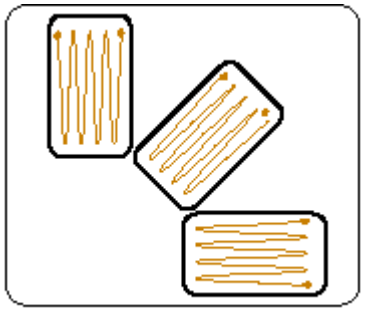
Typical Gauge Patterns: Planer
Rosette Strain Gauges

- There are two main layouts in multi-axial strain rosettes:
  - Planer
  - Stacked



0°-60°-120° Delta Rosette

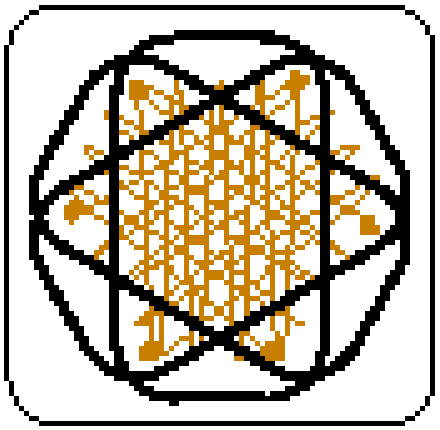
- Select a strain rosette with a planer layout if :
  - Heat dissipation is an issue
  - Accuracy and stability are critical



0°-45°-90° Retangular Rosette

- Planer layout allows each gauge to be closer to the measurement surface and there is no interference between them

Typical Gauge Patterns: Stacked Rosette Strain Gauges

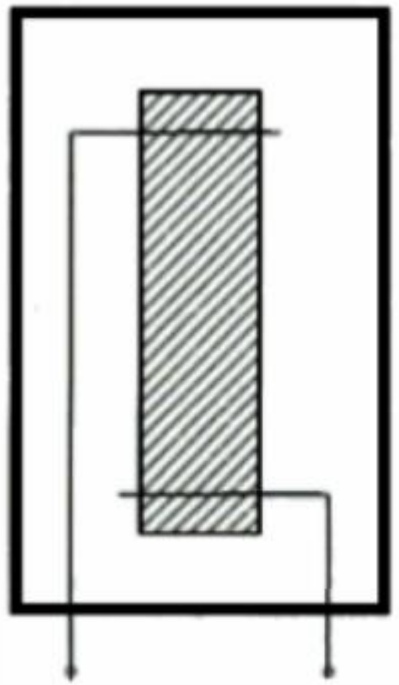


Stacked 0°-60°-120° Delta Rosette

- Select a strain rosette with a stacked layout if :
  - The strain gradient is very large. Stacked layouts measure at the same point
  - Space for mounting the gauge is limited

Semiconductor Strain Gauges

- This uses a piezo-resistive effect. Strain applied make a change in resistivity
- Materials used are normally germanium and silicon
- Advantages
  - High gauge factor
  - Low hysteresis
  - Very small in size
- Disadvantages
  - Very sensitive to changes in temperature
  - Not linear
  - Very expensive



Desirable Characteristics of Strain Gauges

- High gauge factor
- Low temperature coefficient or High resistance temperature coefficient
- Linear [strain vs resistance]
- High resistance

Resistor Combinations

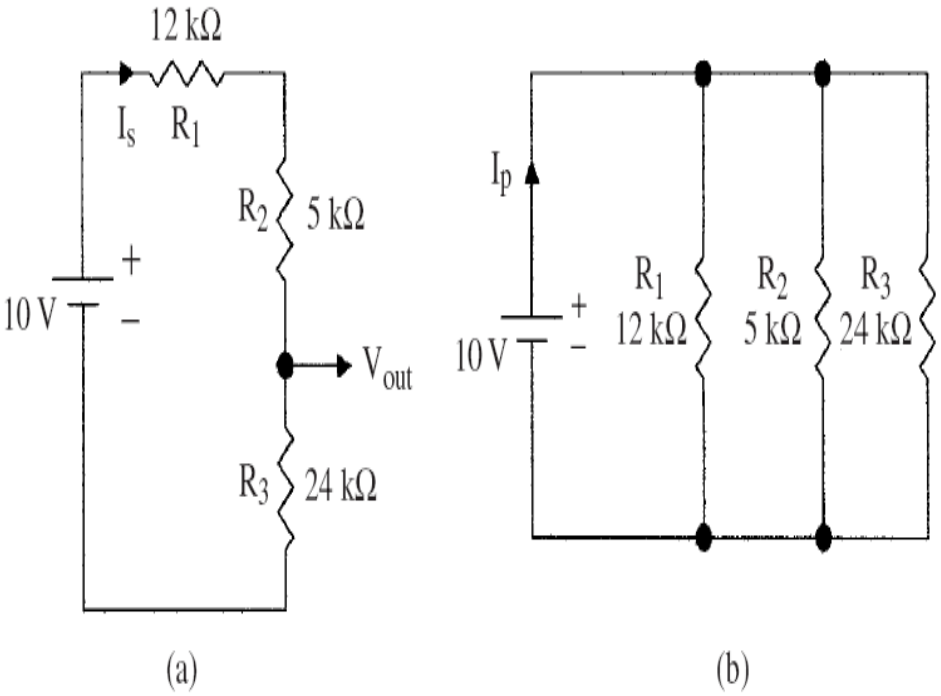
- Series Combination

$$R_T = R_1 + R_2 + R_3 + \ldots$$

- Parallel Combination

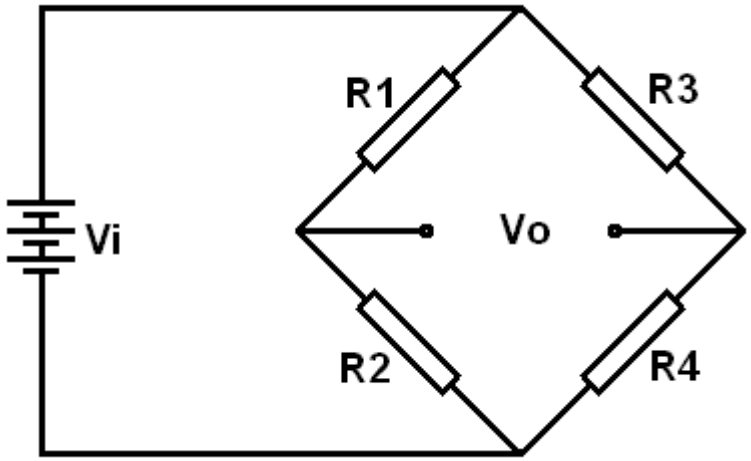
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots$$

- Wheatstone Bridge
- Resistor Example

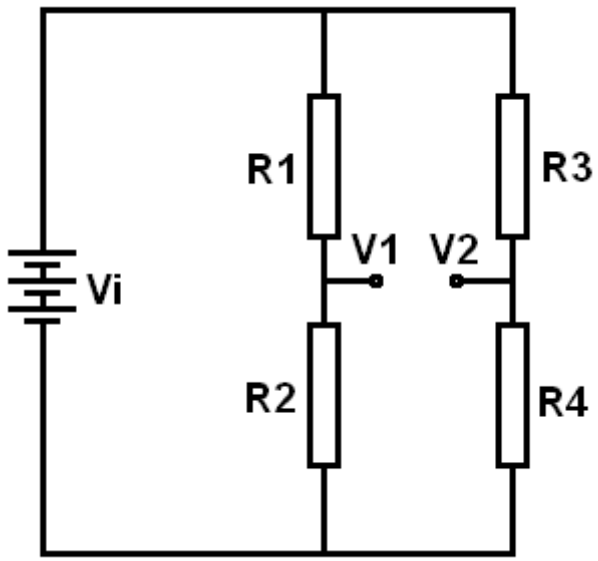


The Wheatstone Bridge

- Contains four resistive elements
- Used to measure small change in resistance
- Used widely in instrumentation to balance out effects such as thermal changes in resistivity
- Used as a differential measurement system



Wheatstone Bridge Equations



$$V_1 = V_i \frac{R_2}{R_1 + R_2} \qquad V_2 = V_i \frac{R_4}{R_3 + R_4}$$

$$V_o = V_1 - V_2$$

$$= \left(V_i \frac{R_2}{R_1 + R_2} \right) - \left(V_i \frac{R_4}{R_3 + R_4} \right)$$

$$= V_i \left[\frac{R_2(R_3 + R_4) - R_4(R_1 + R_2)}{(R_1 + R_2)(R_3 + R_4)} \right]$$

$$= V_i \left[\frac{R_2 R_3 + R_2 R_4 - R_4 R_1 - R_4 R_2}{(R_1 + R_2)(R_3 + R_4)} \right]$$

$$= V_i \left[\frac{R_2 R_3 - R_4 R_1}{(R_1 + R_2)(R_3 + R_4)} \right]$$

Strain Gauges in a Wheatstone Bridge Arrangement

$$V_o = V_i\left[\frac{R_2R_3 - R_4R_1}{(R_1+R_2)(R_3+R_4)}\right]$$

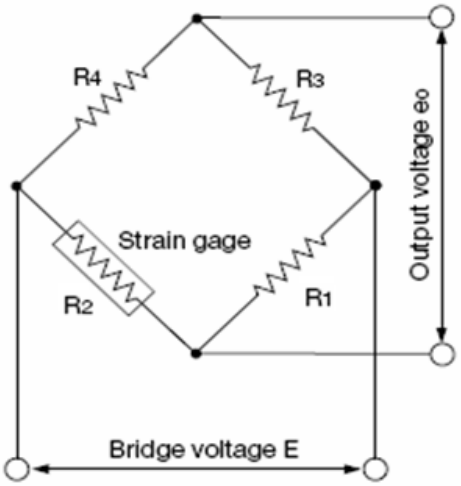
$$= V_i\left[\frac{(R_2+\Delta R)R_3 - R_4R_1}{(R_1+R_2+\Delta R)(R_3+R_4)}\right]$$

$$If\ \ R_1 = R_2 = R_3 = R_4$$

$$V_o = V_i\left[\frac{R^2 + R\Delta R - R^2}{(2R+\Delta R)2R}\right]$$

$$= V_i\left[\frac{R\Delta R}{(4R^2+2R\Delta R)}\right]$$

$$= V_i\left[\frac{\Delta R}{(4R+2\Delta R)}\right]$$



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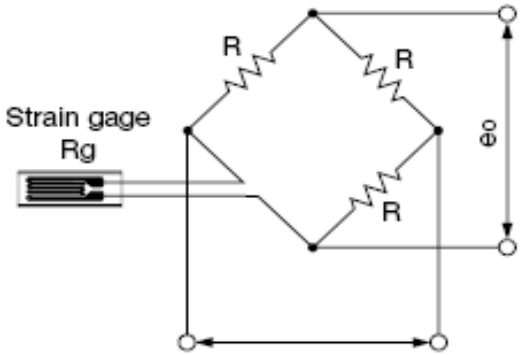
$$R >> \Delta R$$

$$V_o = \frac{1}{4}\cdot\frac{\Delta R}{R}V_i = \frac{V_i}{4}S\varepsilon$$

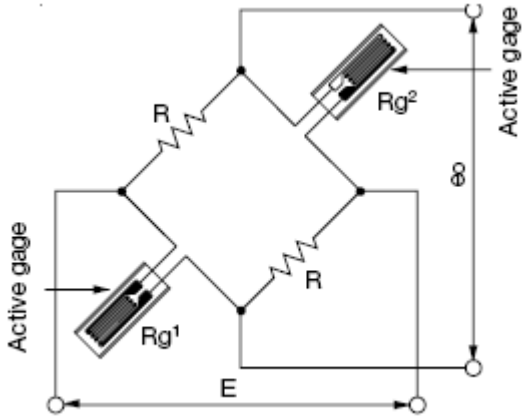
From slide 11

Strain Gauge Arrangements

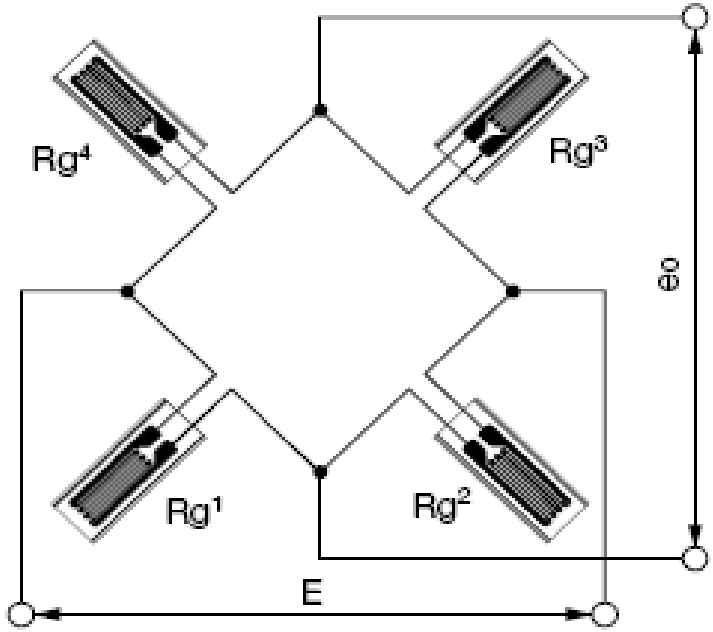
QUARTER BRIDGE



HALF BRIDGE



FULL BRIDGE



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